

Software Requirements Specification (SRS)

for

Banking Core Transaction Processing System (BCTPS)

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Software Requirements Specification

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the requirements for the Banking Core Transaction Processing System (BCTPS). This document serves as the foundation for design, development, testing, and deployment. It defines the functional capabilities, external interfaces, performance benchmarks, security constraints, and regulatory compliance requirements necessary to deliver a reliable, secure, and high-performance banking platform.

1.2 Document Conventions

- Must: Mandatory requirement essential for operation.
- Should: Recommended requirement that adds value but is not critical.
- May: Optional requirement.
- Shall: Binding contractual requirement.
- R#: Unique requirement identifier for traceability.

1.3 Intended Audience

- Development Team — implement the system.
- Project Managers — plan timelines, resources, and scope.
- Quality Assurance Team — create test plans and validation protocols.
- Banking Operations Staff — validate daily operational needs.
- Compliance Officers — ensure PCI-DSS, GDPR, and local banking compliance.
- System Administrators — understand deployment, monitoring, and recovery.

1.4 Product Scope

The BCTPS is a cloud-native, microservices-based platform serving as the central transaction-processing platform for a mid-sized regional bank. It manages customer accounts; processes deposits, withdrawals, transfers, bill payments, and loan repayments; enforces tiered authorization; performs real-time fraud detection; generates statements; and integrates with ATM networks, payment gateways, SWIFT, and regulatory systems.

The system aims to process 500 TPS, achieve 99.999% uptime, and reduce fraud-related losses by 40%.

1.5 References

- IEEE Std 830-1998.
- PCI-DSS v3.2.1.
- GDPR Regulation (EU) 2016/679.
- Basel III Banking Regulations.
- SWIFT MT/MX Message Format Standards.
- System Business Requirements Document (BRD) v2.0.

2. Overall Description

2.1 User Needs

User Type	Needs Description
Bank Customers	Mobile/web access to balances, transfers, bill payments, statements, and transaction alerts.
Bank Tellers	Desktop processing of deposits, withdrawals, and cashier's checks.
Branch Managers	Branch-volume monitoring, high-value transaction approvals, and reconciliation reports.
Fraud Analysts	Investigation dashboard for flagged transactions, behavior analysis, and blocking/approval.
Customer Support Agents	Customer 360 view for account details, transaction history, and disputes.
System Administrators	User management, configuration, health monitoring, and backup/recovery.

2.2 Assumptions and Dependencies

2.2.1 Assumptions

- A-01: Customers have valid government ID and Tax ID for KYC.
- A-02: Existing CRM supplies customer data.
- A-03: Agreements exist with Visa, Mastercard, and SWIFT networks.
- A-04: Transactions follow banking hours and clearing cycles unless real-time.
- A-05: Branches connect through a secure dedicated MPLS network.

2.2.2 Dependencies

- D-01: SWIFT Alliance Gateway for international wire messages.
- D-02: AWS Cloud Platform (EC2, RDS, DynamoDB, S3, KMS, Lambda).
- D-03: Feedzai fraud-detection engine for real-time transaction scoring.
- D-04: Legacy AS/400 Core Banking System for nightly batch and GL updates.
- D-05: Salesforce CRM integration, subject to API credentials and data mapping.

3. System Features and Requirements

3.1 Functional Requirements

3.1.1 Customer & Account Management

- R-101: Tellers/Admins can create customer profiles with identity, contact, employment, and KYC details.
- R-102: Customers can open Checking, Savings, Fixed Deposit, Loan, and Credit Card accounts.
- R-103: Generate Customer IDs CUST-YYYY-XXXXX and Account Numbers ACC-XXXXXXXXXX.
- R-104: Customers can update profile data via mobile/web with OTP verification.
- R-105: Customers can view real-time balance, available balance, and at least 12 months of transaction history.

3.1.2 Transaction Processing (Real-Time)

- R-201: Process cash/check deposits and immediately update balances.
- R-202: Process withdrawals only when available balance is sufficient.
- R-203: Process internal transfers in real time.
- R-204: Process domestic and international external transfers via SWIFT and clearing networks.
- R-205: Process bill payments to registered payees.
- R-206: Generate unique Transaction IDs TXN-YYYYMMDD-XXXXXX.
- R-207: Place holds on transactions requiring additional verification, including amounts above \$10,000.
- R-208: Enforce minimum balances and applicable penalty fees.

3.1.3 Transaction Authorization & Workflow

- R-301: Tier 1 < \$5,000 auto-approved; Tier 2 \$5,000–\$25,000 requires Branch Manager dual authorization; Tier 3 > \$25,000 requires Regional Manager and Compliance Officer approval.
- R-302: Provide Approvals dashboard for pending requests.
- R-303: Notify approvers by push and email.
- R-304: Automatically reject requests pending over 24 hours.
- R-305: Log approval decisions, approver ID, timestamp, and rejection reason.

3.1.4 Fraud Detection & Prevention

- R-401: Analyze every transaction through Feedzai in real time.
- R-402: Assign a 0–100 Risk Score.
- R-403: Automatically block Risk Score > 90 and flag for Fraud Analyst review.
- R-404: Send SMS to customers when Risk Score > 70 asking for confirmation.
- R-405: Allow analysts to permanently block confirmed fraudulent cards/accounts.

- R-406: Maintain blacklists of fraudulent accounts and IP addresses.

3.1.5 Account Statements & Reporting

- R-501: Generate daily statements with opening/closing balances.
- R-502: Generate monthly/annual PDF statements with digital signatures.
- R-503: Allow PDF/CSV downloads for date ranges up to 12 months.
- R-504: Generate regulatory reports such as CTRs for submission to the FIU.

3.1.6 Daily End-of-Day (EOD) Batch Processing

- R-601: Run nightly EOD at 11:59 PM.
- R-602: Calculate/apply daily savings and loan interest.
- R-603: Charge maintenance, overdraft, and late-payment fees.
- R-604: Reconcile transactions against the General Ledger and flag discrepancies.
- R-605: Generate an EOD Reconciliation Report.
- R-606: Roll forward to the next business day after successful EOD.

3.1.7 Audit & Compliance

- R-701: Maintain immutable audit trails for transactions, user actions, account modifications, and authorization decisions for at least 7 years.
- R-702: Never store full credit-card PANs or CVVs; store tokens only.
- R-703: Support GDPR data export and deletion requests subject to legal retention.
- R-704: Allow compliance officers to run ad-hoc transaction queries for reporting and investigations.

3.2 External Interface Requirements

3.2.1 Hardware Interfaces

- HI-01: Interface with ATM switches via ISO 8583 for withdrawals and balance inquiries.
- HI-02: Interface with POS terminals through the payment gateway for card transactions.

3.2.2 Software Interfaces (APIs)

- SI-01: Integrate Stripe/Adyen via REST for online card and digital-wallet transactions.
- SI-02: Integrate SWIFT Alliance Gateway using MT/MX and ISO 20022.
- SI-03: Integrate domestic ACH/RTGS networks via SFTP or REST.
- SI-04: Send transaction data to Feedzai REST API and receive risk score within 200ms.
- SI-05: Send GL posting files to AS/400 via SFTP every 30 minutes.
- SI-06: Synchronize customer profile data with Salesforce via REST API.

3.2.3 User Interfaces (UI)

- UI-01: Customer app prominently shows current balance and five recent transactions.
- UI-02: Teller desktop provides Quick Action panel for deposit, withdrawal, and transfer.
- UI-03: Approvals dashboard sorts pending requests by highest amount and provides Approve/Reject actions.
- UI-04: Fraud screen shows customer transaction timeline, geolocation, device fingerprint, and risk score.

3.3 Non-Functional Requirements

3.3.1 Performance & Scalability

- NFR-01: Process 500 TPS during peak hours.
- NFR-02: Support 5 million customer accounts and 10 million transactions/day.
- NFR-03: Load balances and transaction history in under 500ms.
- NFR-04: Fraud risk score returned within 200ms.
- NFR-05: EOD process completes within 2 hours by 2:00 AM.

3.3.2 Reliability & Availability

- NFR-06: Guarantee 99.999% annual uptime (~5.26 minutes downtime).
- NFR-07: Multi-Active deployment across 3 AWS Availability Zones.
- NFR-08: PostgreSQL backups every hour with PITR.
- NFR-09: Database failover within 30 seconds.
- NFR-10: Circuit Breaker pattern for payment, SWIFT, and fraud integrations.

3.3.3 Security & Compliance

- NFR-11: TLS 1.3 for data in transit.
- NFR-12: AES-256 for sensitive data at rest.
- NFR-13: HMAC or digital signatures for transaction integrity and non-repudiation.
- NFR-14: Session timeout of 15 minutes for web and 30 minutes for mobile.
- NFR-15: Immutable audit logs in a separate write-once database.
- NFR-16: RBAC using least privilege.

3.3.4 Usability

- NFR-17: Material Design principles for the customer app.
- NFR-18: FaceID/TouchID/Fingerprint authentication.
- NFR-19: Keyboard-friendly teller workflow.
- NFR-20: Clear actionable errors instead of technical codes.
- NFR-21: English/Spanish localization and regional date/time formats.

3.3.5 Maintainability

- NFR-22: Docker and Kubernetes with rolling updates and canary deployments.
- NFR-23: Minimum 90% unit-test coverage plus external API integration tests.
- NFR-24: JSON logs aggregated to centralized ELK.
- NFR-25: Expose /health, /ready, /live and Prometheus metrics for monitoring.

4. Appendixes

Appendix A: Data Dictionary

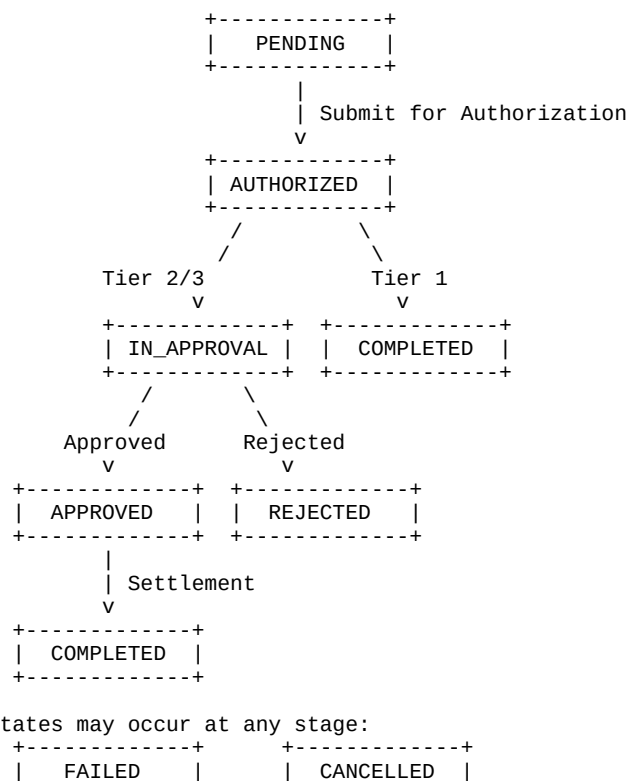
Data Element	Data Type	Description	Constraints
Customer_ID	VARCHAR(20)	Unique customer identifier.	Primary Key; CUST-YYYY-XXXXX.
Account_Number	VARCHAR(20)	Unique bank account identifier.	Primary Key; ACC-XXXXXXXXXX.
Account_Type	ENUM	Type of bank account.	CHECKING, SAVINGS, FIXED_DEPOSIT, LOAN, CREDIT_CARD.
Available_Balance	DECIMAL(15,2)	Balance available for withdrawal excluding holds.	Must be >= 0.
Transaction_ID	VARCHAR(20)	Unique transaction identifier.	Primary Key; TXN-YYYYMMDD-XXXXXX.
Transaction_Type	ENUM	Type of transaction.	DEPOSIT, WITHDRAWAL, INTERNAL_TRANSFER, EXTERNAL_TRANSFER, BILL_PAYMENT, LOAN_REPAYMENT.
Transaction_Status	ENUM	Current transaction status.	PENDING, AUTHORIZED, APPROVED, REJECTED, FAILED, COMPLETED.
Risk_Score	INTEGER (0-100)	Fraud risk score.	Calculated field.
Authorization_Tier	INTEGER	Required authorization level.	1 (<\$5k), 2 (\$5k-\$25k), 3 (>\$25k).
Approver_ID	VARCHAR(20)	Approver/rejector user ID.	Nullable; mandatory for Tier 2/3.
Audit_Trail	JSON	Immutable user-action/change log.	Separate immutable database.

Appendix B: Acronyms and Abbreviations

Acronym	Full Form
BCTPS	Banking Core Transaction Processing System
PCI-DSS	Payment Card Industry Data Security Standard
GDPR	General Data Protection Regulation
KYC	Know Your Customer
AML	Anti-Money Laundering
SWIFT	Society for Worldwide Interbank Financial Telecommunication
ACH	Automated Clearing House
RTGS	Real-Time Gross Settlement
FedWire	Federal Reserve Wire Network
NACHA	National Automated Clearing House Association
FIU	Financial Intelligence Unit
CTR	Currency Transaction Report
GL	General Ledger
EOD	End of Day
PITR	Point-In-Time Recovery
TPS	Transactions Per Second
SLA	Service Level Agreement
K8s	Kubernetes
ELK	Elasticsearch, Logstash, Kibana
HMAC	Hash-based Message Authentication Code
CBS	Core Banking System (Legacy)
CRM	Customer Relationship Management

Appendix C: Transaction State Machine

This appendix defines the valid state transitions for a transaction in the BCTPS.



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State Transition Rules

- A transaction in PENDING can move to AUTHORIZED, FAILED, or CANCELLED.
- A transaction in AUTHORIZED can move to IN_APPROVAL (Tier 2/3) or COMPLETED (Tier 1).
- A transaction in IN_APPROVAL can move to APPROVED, REJECTED, or FAILED.
- A transaction in APPROVED must move to COMPLETED (settlement).
- Once COMPLETED, a transaction cannot be modified; it can only be reversed through a separate reversal transaction.

End of Document